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ON THE ACOUSTIC BOUNDARY CONDITION IN THE PRESENCE OF FLOW

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The boundary condition on the acoustic perturbation velocity at an impermeable surface in a flow is considered for the cases in which the surface generates a sound field by vibration or is acoustically deformed by an incident sound field. It is shown that in general the condition is not equivalent to the requirement of continuity of acoustic particle displacement in the direction normal to the unperturbed surface.

1. INTRODUCTION

In a recent paper [1], Taylor has determined the acoustic field of a vibrating sphere in the presence of a steady potential flow at low Mach number. Included in this work is a derivation of the appropriate boundary condition to be applied to the unsteady perturbations of the base flow on the surface of a vibrating solid body when the problem is considered in a frame of reference in which the body generates a sound field by small motions about a steady mean position. It is the purpose of the present paper to present a somewhat more formal derivation of the same boundary condition which the author has found to be useful. The more formal approach has certain advantages, perhaps the primary one being that it allows straightforward manipulation of the condition into a form which is more convenient to apply than is that of reference [1]. In addition it permits a simple generalization to describe a passive boundary on which the normal acoustic impedance is known.

There has been considerable discussion, and even some controversy, in the literature during the past two decades concerning the proper boundary conditions on the acoustic field in the presence of a base flow. Possibly the first correct accounting for the effect of a moving medium was given almost concurrently by Miles [2] and Ribner [3] for a plane interface of relative motion. Ingard [4] later determined the effect of flow on the boundary condition at a plane impedance surface. Especially in the area of duct acoustics, a great deal of discussion has followed in the literature over the apparent differences in the condition depending upon whether the moving acoustic medium is treated as viscous or inviscid. A comprehensive review of the situation appears in the work of Nayfeh, Kaiser and Telionis [5], where numerous references are cited.

As a result of this discussion it has gradually been accepted that when the base flow is treated as inviscid the proper kinematic condition at an impenetrable boundary of the acoustic field is expressed physically by the requirement of "continuity of acoustic particle displacement". This means specifically that at any such surface which generates an acoustic field by small deformations, or undergoes small deformations in response to an acoustic field, the acoustic fluid particle displacement in the direction normal to the undeformed boundary must equal the displacement of the boundary itself in the same direction.

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It will appear in the results to follow, and was suggested also in reference [1], that the above physical statement of the boundary condition is incorrect in general. It is valid for the cases treated in references [2–4] owing to the fact that in these the authors considered (a) only surfaces whose undeformed shape is planar, and (b) only base flows which are everywhere parallel to those planes. When more general surface shapes or base flows are considered the terms in the boundary condition which can be overlooked when using the physical statement above are significant, as was demonstrated in the examples of reference [1].

2. FORMULATION

Consider a surface $S(t)$ which moves relative to a given reference frame. The surface is assumed to be in contact with a fluid whose Eulerian velocity relative to the frame of reference is $\bar{\mathbf{u}}(\mathbf{x}, t)$. Then, within the continuum theory of motion of the fluid, the conditions which must be satisfied at each point of the moving surface, assuming it is impenetrable to the fluid, can be stated as

$$\bar{\mathbf{u}} \cdot \boldsymbol{\nu} = \mathbf{V} \cdot \boldsymbol{\nu}, \quad \bar{\mathbf{u}} - (\bar{\mathbf{u}} \cdot \boldsymbol{\nu})\boldsymbol{\nu} = \mathbf{V} - (\mathbf{V} \cdot \boldsymbol{\nu})\boldsymbol{\nu} \tag{1a, b}$$

on $S(t)$. In equation (1), $\mathbf{V}(\mathbf{x}, t)$ is the Eulerian expression for the velocity of the surface $S(t)$ relative to the frame of reference, and $\boldsymbol{\nu}(t)$ is the instantaneous unit normal vector to the surface. Condition (1a) is the kinematic condition expressing the fact that no voids occur at the interface between $S(t)$ and the fluid, while condition (1b) is the no-slip requirement which applies to a viscous fluid. If the fluid is considered to be inviscid then the second condition is dropped. It is condition (1a) which will be of primary concern in the present work.

If one assumes that the surface S is described spatially by a relation

$$f(\mathbf{x}, t) = 0, \tag{2}$$

then, provided that f is defined so that it is positive on the fluid side of S and negative on the other, the normal vector $\boldsymbol{\nu}(t)$ is given by

$$\boldsymbol{\nu}(t) = \nabla f / |\nabla f| \qquad \text{on } f = 0, \tag{3}$$

and $\boldsymbol{\nu}(t)$ always points into the fluid. Because a point $\mathbf{x}$ on S remains on S for all time, equation (2) can be differentiated with respect to t to obtain $\nabla f \cdot (\mathrm{d}\mathbf{x}/\mathrm{d}t) + \partial f/\partial t = 0$ on $f = 0$, from which using equation (3), one obtains the well-known relation

$$\mathbf{V} \cdot \boldsymbol{\nu} = -(1/|\nabla f|)\, \partial f/\partial t \qquad \text{on } f = 0. \tag{4}$$

Thus condition (1a) can be rewritten by using equation (4) as

$$\bar{\mathbf{u}} \cdot \nabla f = -\partial f/\partial t \qquad \text{on } f = 0. \tag{5}$$

It is assumed that the surface S either generates a sound field in the fluid by vibration or is acoustically deformed in response to an incident sound field from the fluid. In this case it is convenient to choose the reference frame for $\mathbf{V}$ and $\bar{\mathbf{u}}$ such that the motion of S is a small perturbation about a stationary mean surface S_0, and the corresponding fluid velocity field is a small perturbation about a steady base flow $\mathbf{U}(\mathbf{x})$. To describe this situation it proves to be of value to introduce temporarily a local orthogonal curvilinear co-ordinate system $\alpha(\mathbf{x})$, $\beta(\mathbf{x})$, $\gamma(\mathbf{x})$, fixed on the surface S_0 in such a way that $\alpha(\mathbf{x}) = 0$ on S_0 and $\nabla\alpha \cdot \nabla\beta = \nabla\beta \cdot \nabla\gamma = \nabla\gamma \cdot \nabla\alpha = 0$. That is, the co-ordinate α is normal to S_0 while β and γ are tangential co-ordinates lying in S_0. Then $S(t)$ can be described by $\alpha = \varepsilon g(\beta, \gamma, t) + O(\varepsilon^2)$, while the fluid velocity is of the form $\bar{\mathbf{u}}(\mathbf{x}, t) = \mathbf{U}(\mathbf{x}) + \varepsilon\mathbf{u}(\mathbf{x}, t) + O(\varepsilon^2)$, in which ε is a dimensionless parameter characterizing the magnitude of the acoustic disturbances.

In terms of the curvilinear co-ordinates, relation (2) is $\alpha - \varepsilon g + \cdots = 0$ so that the kinematic boundary condition (5) can be written

$$(\mathbf{U} + \varepsilon\mathbf{u} + \cdots) \cdot \nabla(\alpha - \varepsilon g + \cdots) - \varepsilon(\partial/\partial t)(g + \cdots) = 0 \qquad \text{on } \alpha = \varepsilon g + \ldots. \tag{6}$$

Condition (5) is satisfied, of course, by the steady base flow in the absence of the acoustic perturbation. This is expressed by setting $\varepsilon = 0$ in equation (6):

$$\mathbf{U} \cdot \nabla\alpha = 0 \qquad \text{on } \alpha = 0. \tag{7}$$

That is, the steady base flow $\mathbf{U}$ must be tangent to the undeformed stationary surface S_0. In the following section, the form taken by equation (6) for $\varepsilon \ll 1$ will be examined.

3. LINEARIZATION OF THE BOUNDARY CONDITION

If the functions $\mathbf{U}$ and $\mathbf{u}$ are temporarily regarded as functions of the independent variables α, β, γ, then, provided they are sufficiently smooth at $\alpha = 0$, equation (6) can be expanded in powers of ε. For example,

$$\mathbf{U} \cdot \nabla\alpha|_{\alpha=\varepsilon g+\cdots} = \mathbf{U} \cdot \nabla\alpha|_{\alpha=0} + \varepsilon g[(\partial/\partial\alpha)(\mathbf{U} \cdot \nabla\alpha)]|_{\alpha=0} + O(\varepsilon^2),$$

$$\varepsilon\mathbf{u} \cdot \nabla\alpha|_{\alpha=\varepsilon g+\cdots} = \varepsilon\mathbf{u} \cdot \nabla\alpha|_{\alpha=0} + O(\varepsilon^2),$$

and so on. Thus, using equation (7), one can put condition (6) in the form

$$\varepsilon[g(\partial/\partial\alpha)(\mathbf{U} \cdot \nabla\alpha) + \mathbf{u} \cdot \nabla\alpha - \mathbf{U} \cdot \nabla g - \partial g/\partial t]|_{\alpha=0} + O(\varepsilon^2) = 0.$$

To leading order, for $\varepsilon \ll 1$, this implies that

$$\mathbf{u} \cdot \nabla\alpha = (\partial/\partial t + \mathbf{U} \cdot \nabla)g - g(\partial/\partial\alpha)(\mathbf{U} \cdot \nabla\alpha) \qquad \text{on } \alpha = 0. \tag{8}$$

Equation (8) is the linearized boundary condition governing the acoustic perturbation velocity $\mathbf{u}$ at an impermeable surface. As indicated, it is to be applied on the mean position, S_0, of the moving surface as is usual in linearized acoustics. It is a simple matter to show that it is equivalent to the condition given in reference [1] when account is taken of notational differences. However, the derivation of condition (8) in terms of the orthogonal co-ordinates α, β, γ makes it possible to manipulate the expression rather easily into a form which is perhaps more convenient to interpret and to apply. For this purpose it is necessary to employ several well-known relations from the theory of orthogonal curvilinear co-ordinates. The reader is referred to reference [6], Appendix 2, for example, where a concise summary of this theory is given.

The set of unit base vectors along the co-ordinate lines α, β, γ are here denoted as $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, respectively, with corresponding scale factors h_α, h_β, h_γ. If the relation [6] $\nabla\alpha = |\nabla\alpha|\mathbf{a} = (1/h_\alpha)\mathbf{a}$ is used, equation (8) can be written as

$$\mathbf{u} \cdot \mathbf{a} = h_\alpha(\partial g/\partial t + \mathbf{U} \cdot \nabla g) - g(\mathbf{a} \cdot \partial\mathbf{U}/\partial\alpha + \mathbf{U} \cdot \partial\mathbf{a}/\partial\alpha) \qquad \text{on } \alpha = 0,$$

or, after bringing h_α inside the material derivative, as

$$\mathbf{u} \cdot \mathbf{a} = (\partial/\partial t + \mathbf{U} \cdot \nabla)(g h_\alpha) - g[\mathbf{a} \cdot \partial\mathbf{U}/\partial\alpha + \mathbf{U} \cdot (\partial\mathbf{a}/\partial\alpha + \nabla h_\alpha)] \qquad \text{on } \alpha = 0. \tag{9}$$

One now introduces the relations [6]

$$\frac{\partial\mathbf{a}}{\partial\alpha} = -\frac{1}{h_\beta}\frac{\partial h_\alpha}{\partial\beta}\mathbf{b} - \frac{1}{h_\gamma}\frac{\partial h_\alpha}{\partial\gamma}\mathbf{c}, \quad \nabla h_\alpha = \frac{1}{h_\alpha}\frac{\partial h_\alpha}{\partial\alpha}\mathbf{a} + \frac{1}{h_\beta}\frac{\partial h_\alpha}{\partial\beta}\mathbf{b} + \frac{1}{h_\gamma}\frac{\partial h_\alpha}{\partial\gamma}\mathbf{c}$$

into the last two terms in equation (9), which become

$$-g\mathbf{U}\,.\,(1/h_\alpha)(\partial h_\alpha/\partial\alpha)\mathbf{a} = -g(\partial h_\alpha/\partial\alpha)\mathbf{U}\,.\,\nabla\alpha.$$

This vanishes on $\alpha = 0$ on account of equation (7) so that condition (9) is

$$\mathbf{u}\,.\,\mathbf{a} = (\partial/\partial t + \mathbf{U}\,.\,\nabla)(gh_\alpha) - g\mathbf{a}\,.\,\partial\mathbf{U}/\partial\alpha \qquad \text{on } \alpha = 0. \tag{10}$$

Finally it can be noted that $\partial\mathbf{U}/\partial\alpha = h_\alpha\mathbf{a}\,.\,\nabla\mathbf{U}$, and that on the surface $\alpha = 0$ the unit vector $\mathbf{a}$ is simply the unit normal to S_0 directed into the fluid. This vector can be denoted as $\mathbf{n}$. Then the boundary condition (10) becomes

$$\mathbf{u}\,.\,\mathbf{n} = (\partial/\partial t + \mathbf{U}\,.\,\nabla)(g/|\nabla\alpha|) - (g/|\nabla\alpha|)\mathbf{n}\,.\,(\mathbf{n}\,.\,\nabla\mathbf{U}) \qquad \text{on } S_0. \tag{11}$$

Condition (11) is the major result of the present paper. The boundary condition is expressed without specific reference to the curvilinear co-ordinate system. The mean surface S_0 is given by $\alpha(\mathbf{x}) = 0$, and g may be considered now as a function of $\mathbf{x}$ and t. These functions, as well as the normal vector $\mathbf{n}$ are known if S_0 and its perturbed motion are specified. This would be the case, for example, if S_0 were a solid surface generating sound by small vibrations. The condition (11) also applies on either side of an interface between different acoustic media. In that case, the right-hand side of equation (11) would be evaluated on each side of the interface given by $\alpha = 0$, and the two expressions equated to ensure continuity of acoustic normal velocity (in the direction $\boldsymbol{\nu}$) across the interface. Of course in this circumstance there is also required a condition expressing continuity of pressure across the interface. This second condition would be used together with equation (11) to eliminate the unknown function g from the problem.

Because of the form of condition (11) it is also a simple matter to derive the boundary condition appropriate to an acoustically absorbing boundary whose normal impedance z for harmonic motion is known. Within the context of the present work this derivation involves determining the function $g/|\nabla\alpha|$ which appears in equation (11) so that it represents the acoustic deformation of such a boundary. Consider the expression

$$(\bar{p} - p_0)/\mathbf{V}\,.\,\boldsymbol{\nu} \tag{12}$$

where $\bar{p}$ is the total fluid pressure, p_0 the ambient pressure, and $\mathbf{V}\,.\,\boldsymbol{\nu}$ the surface normal velocity as before. This can be written

$$\frac{(\varepsilon p + \cdots)|\nabla\alpha - \varepsilon\nabla g + \cdots|}{\varepsilon\,\partial g/\partial t + \cdots} = \frac{(|\nabla\alpha|p)_{\alpha=0}}{\partial g/\partial t} + O(\varepsilon). \tag{13}$$

For harmonic time dependence, the complex acoustic pressure p and g are assumed proportional to $\exp(\mathrm{i}\omega t)$ so that to leading order in ε expression (13) is $(1/\mathrm{i}\omega g)(|\nabla\alpha|p)_{\alpha=0}$. This leading approximation to expression (12) is, of course, minus the complex normal impedance, z, of the surface (note that $\boldsymbol{\nu}$ is defined so that it points away from the surface). Thus for the impedance boundary one has

$$g = -(1/\mathrm{i}\omega z)(|\nabla\alpha|p)_{\alpha=0}, \tag{14}$$

and the boundary condition (11) becomes

$$\mathbf{u}\,.\,\mathbf{n} = -(p/z) - (1/\mathrm{i}\omega)\mathbf{U}\,.\,\nabla(p/z) + (p/\mathrm{i}\omega z)\mathbf{n}\,.\,(\mathbf{n}\,.\,\nabla\mathbf{U}) \qquad \text{on } S_0. \tag{15}$$

The second term in equation (15) has been written as though the differentiation were to be carried out before setting $\alpha = 0$, contrary to what is stated by equation (14). This makes no difference, however, since the α-derivative term in $(\mathbf{U}\,.\,\nabla)$ is multiplied by the component of $\mathbf{U}$ in the α-direction, which vanishes on S_0 by equation (7).

4. DISCUSSION

Since fluid velocity is the basic kinematic quantity in the Eulerian description of fluid motion, it is important to emphasize that condition (11), however it may be interpreted in special cases, expresses the linearized version of the requirement of continuity of normal velocity on $S(t)$ as stated in condition (1a). This is true regardless of whether the base flow is or is not required to satisfy the no-slip condition. By using equation (4) it can be seen that the first term on the right side of condition (11) is the first approximation to the normal velocity (in the direction $\boldsymbol{\nu}$) of a point on the surface S. The two terms involving the flow velocity $\mathbf{U}$ arise from the linearization process, which expresses the condition in terms of $\mathbf{n}$ and S_0 rather than in terms of $\boldsymbol{\nu}$ and S. As can be seen from equation (6), the term involving $\mathbf{U}\cdot\nabla g$ appears because the instantaneous normal vector $\boldsymbol{\nu}$ is inclined relative to $\mathbf{n}$ by an angle whose magnitude is of order ε. Thus the flow velocity $\mathbf{U}$ has a component in the direction of $\boldsymbol{\nu}$ which is of order ε and which adds to the acoustic velocity in forming the total perturbation velocity normal to S. The third term in condition (11) is a consequence of the fact that the flow velocity $\mathbf{U}$ at the actual location of the point on S differs from its value at the image of the point on S_0 by an amount which is also of order ε in general. Thus the total unsteady perturbation velocity normal to S is, to the first approximation,

$$\varepsilon[\mathbf{u}\cdot\mathbf{n}-\mathbf{U}\cdot\nabla(g/|\nabla\alpha|)+(g/|\nabla\alpha|)\mathbf{n}\cdot(\mathbf{n}\cdot\nabla\mathbf{U})]_{S_0},$$

and condition (11) equates this to the first approximation for the normal velocity of the surface S.

It is, however, of interest in the light of the remarks made in section 1 to consider the implications of condition (11) in terms of fluid particle displacement. From the relation $\alpha=\varepsilon g$, which defines the surface S in the linearized approximation, it follows that the quantity $\varepsilon g/|\nabla\alpha|_{\alpha=0}$ is the leading approximation to the displacement of a point on S_0 in the direction normal to S_0: i.e., along $\mathbf{n}$. This quantity can be denoted by ξ_b. It is convenient to work with the boundary condition in the form of equation (10) where $\mathbf{U}$, $\mathbf{u}$, and g are regarded as functions of the curvilinear co-ordinates α, β, γ. Equation (10) may be written as

$$\mathbf{a}\cdot \mathrm{D}_0\boldsymbol{\xi}/\mathrm{D}t=D_0\xi_b/\mathrm{D}t-(\xi_b/h_\alpha)\mathbf{a}\cdot\partial\mathbf{U}/\partial\alpha \qquad \text{on } \alpha=0, \tag{16}$$

in which $\mathrm{D}_0/\mathrm{D}t\equiv\partial/\partial t+\mathbf{U}\cdot\nabla$, and $\boldsymbol{\xi}$ is the acoustic fluid particle displacement vector defined by $\mathbf{u}=\mathrm{D}_0\boldsymbol{\xi}/\mathrm{D}t$. If this is expressed as

$$\frac{\mathrm{D}_0}{\mathrm{D}t}(\boldsymbol{\xi}\cdot\mathbf{a})-\boldsymbol{\xi}\cdot(\mathbf{U}\cdot\nabla\mathbf{a})=\frac{\mathrm{D}_0\xi_b}{\mathrm{D}t}-\frac{\xi_b}{h_\alpha}\mathbf{a}\cdot\frac{\partial\mathbf{U}}{\partial\alpha} \qquad \text{on } \alpha=0,$$

and if the expressions of reference [6] for the derivatives of the unit vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are employed along with equation (7), condition (16) becomes

$$\begin{aligned}\frac{\mathrm{D}_0}{\mathrm{D}t}(\boldsymbol{\xi}\cdot\mathbf{a})-\boldsymbol{\xi}\cdot\left[\frac{U_\beta}{h_\alpha h_\beta}\frac{\partial h_\beta}{\partial\alpha}\mathbf{b}+\frac{U_\gamma}{h_\alpha h_\gamma}\frac{\partial h_\gamma}{\partial\alpha}\mathbf{c}\right]&\\ =\frac{\mathrm{D}_0\xi_b}{\mathrm{D}t}-\frac{\xi_b}{h_\alpha}\left[\frac{\partial U_\alpha}{\partial\alpha}+\frac{U_\beta}{h_\beta}\frac{\partial h_\alpha}{\partial\beta}+\frac{U_\gamma}{h_\gamma}\frac{\partial h_\alpha}{\partial\gamma}\right]& \qquad \text{on } \alpha=0,\end{aligned} \tag{17}$$

in which U_α, U_β, U_γ are the components of $\mathbf{U}$ along the α, β, γ co-ordinate axes, respectively.

Provided that the base flow is treated as inviscid, then both the second term on the left side and the second term on the right side of equation (17) are non-zero in general. In this case equation (17) illustrates the point made earlier: the boundary condition (11) is not equivalent to a statement of continuity of acoustic particle displacement in the form $\boldsymbol{\xi}\cdot\mathbf{n}=\xi_b$ on S_0.

On the other hand, for a viscous base flow satisfying the no-slip condition U_α, U_β, and U_γ all vanish on $\alpha=0$. In addition, the equation of continuity for the steady flow is

$$\rho\nabla\cdot\mathbf{U}+\mathbf{U}\cdot\nabla\rho=0, \tag{18}$$

where ρ is the fluid density. Hence, $\nabla\cdot\mathbf{U}=0$ on $\alpha=0$. But the tangential derivatives of $\mathbf{U}$ vanish on $\alpha=0$ since $\mathbf{U}$ is everywhere zero there, so that the continuity equation (18) yields [6] $(\partial/\partial\alpha)(h_\beta h_\gamma U_\alpha)=0$ on $\alpha=0$, from which one finds that $\partial U_\alpha/\partial\alpha$ vanishes on $\alpha=0$ as well. When one now returns to conditions (11) and (15) it follows that, for a viscous base flow, both the second and the third terms vanish, leaving the conditions in the form $\mathbf{u}\cdot\mathbf{n}=(1/|\nabla\alpha|)\,\partial g/\partial t$ and $\mathbf{u}\cdot\mathbf{n}=-p/z$ on S_0, which are the standard results.

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